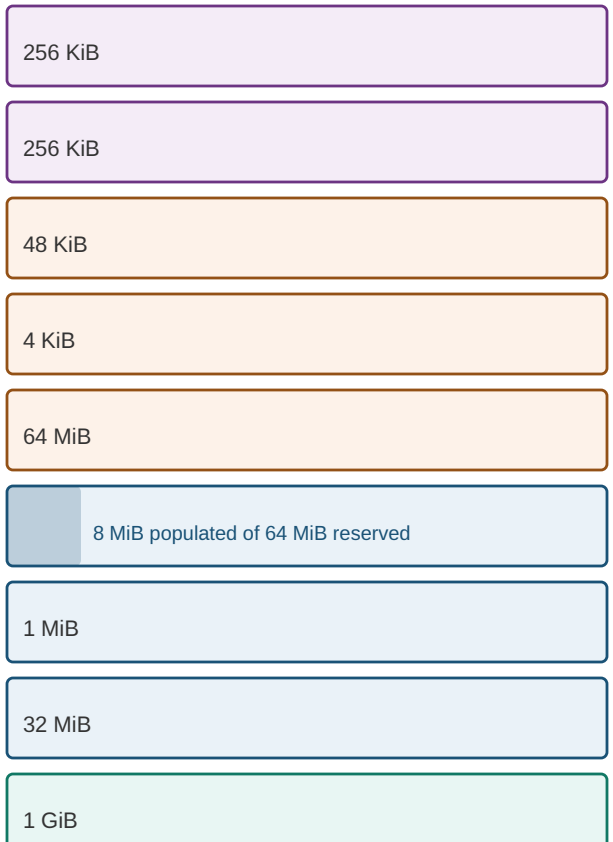
D3 — cva6-multi quadcore CVA6 cluster: address map

GENERATED from sw/addrmap/cva6-multi-map.json. Bands are equal height, not to scale — the map spans 3 GiB and the smallest region is 4 KiB.



0x00000000 0x0003FFFF	**debug**	Cheshire debug ROM
0x02000000 0x0203FFFF	**bootrom**	Cheshire boot ROM. Harts reset here; QEMU needs resetvec set explicitly since it defaults ...
0x02040000 0x0204BFFF	**clint**	SiFive-compatible: MSWI at base, MTIMER at +0x4000, mtime at +0xbff8
0x03002000 0x03002FFF	**uart0**	reg_shift is the property that drifted. Word-spaced, confirmed by a store fault at base+0xC ...
0x04000000 0x07FFFFFF	**plic**	Standard 64 MiB window. Contexts are M then S per hart.
0x20000000 0x23FFFFFF	**scratchpad**	D24 local alias, cacheable half. D26: 8 MiB of the 64 MiB reserved.
0x40000000 0x400FFFFF	**clusterregs**	D24 local alias, non-cacheable half: per-cluster MMIO. Stubbed as RAM; real blocks are ...
0x50000000 0x51FFFFFF	**window**	D23 cluster 0 global aperture, decoded from phase 1 with self-turnaround (D40).
0x80000000 0xBFFFFFFF	**dram**	D22. Size is the model default; -m overrides it.

Cheshire external window 0x20000000–0x7FFFFFFF (D22) — everything the cluster owns sits inside it, or gets no PMA attribute at all.
Cluster N's aperture is 0x50000000 + (N « 25); cluster 0's is decoded from phase 1 and self-turnarounds (D40). mhartid = (cluster « 4) | core (D25).
Shaded portion of a band is backing store; the full band is reserved address space. They differ only for the scratchpad (D26).